

A Bistable, Trigger-induced, Compliant Laparoscopic Grasper - Stage II

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1 Problem Statement

The existing laparoscopic graspers face significant manufacturing and operational challenges, requiring multi-component assemblies with precise tolerance control across components. With rigid, multi-jointed linkages, the current designs lack sufficient force feedback and compliance. Poor mechanical design can lead to joints with friction and a non-linear end effector force relation. Surgeons performing Minimally Invasive Surgeries (MIS) experience more discomfort and pain compared to an open surgery, as MIS can be physically and mentally more demanding. The forces that act on the tissue are distorted due to internal component interactions. The grasping instruments used in minimally invasive surgery reduce the ability of the surgeon to feel the forces applied on the tissue, thereby complicating the handling of the tissue and increasing the risk of tissue damage [1]. We aim to propose and test a fully compliant mechanism, driven by a trigger-induced, bistable V-Beam mechanism which has a significantly lower part count, higher compliance and haptic feedback. The mechanism is primarily aimed at surgeons performing MIS procedures, and its secondary stakeholders include medical device manufacturers aiming to reduce production costs and complexity. Some improvements which can be observed in our design, based on the progress we made, are the reduction in part count, simplification of design and improved ergonomics.

2 Design Constraints

Since this is a medical device innovation, the grasper must be designed within strict dimensional, weight and portability limits with standard trocar sizes of 5-12 mm. We need to ensure that the handheld portion is lightweight (<100g) to minimize surgeon fatigue. We hope to achieve tolerances of $\pm 0.05 - 0.1$ mm on functional dimensions such as the V-Beam length, thickness, travel of bistable actuator and jaw opening as a result of the prototyping, but for an actual laparoscopic grasper, the tolerances should be lesser. The bistable V-Beam must be validated through Finite Element Analysis to ensure permissible force transmission to the compliant grasper, allowing it to deliver grasping forces in the range of about 30-35N, suitable for tissue handling.

3 Review of Existing Solutions

Conventional laparoscopic graspers typically employ either single action or double action jaws actuated by scissor-type handles, and these mechanisms are usually achieved through four-bar linkages, cable-driven systems or rack and pinion systems. While a single action grasper (SAG) has simple mechanisms, it has poor tactile feedback, and a double action grasper (DAG), whose jaws move symmetrically is widely

used in laparoscopy, it has multiple linkages which lead to backlash errors. Conventional graspers include the curved jaw double action graspers such as the Maryland dissector, which has overcome the limitations of SAG & DAG, but can cause tissue damage due to poor feedback. Similarly the Babcock grasper has fenestrated jaws and is used for grasping bowel and appendix tissues. Gear/rack and pinion driven graspers such as Karl Storz precision gear driven dissector, convert rod motion to rotational jaw motion, but face issues with sterilization. The recent advancements in laparoscopic graspers have enabled them to overcome issues in mechanism complexity, improve ergonomics and perform precision grasping, but have evolved as tissue-specific designs with a minimum of 10 parts (moving and non-moving). Robotic innovations such as Stanford Research Center's Da Vinci are known for their precise motion, and tremor filtration, but are still limited by cost and accessibility.

4 Proposed Design - Detailed Description

Our proposed design actively couples a bistable trigger mechanism with a compliant gripping end effector, which is effective over rigid designs for tissue grasping. The compliant trigger coupled with the V-Beam bistable mechanism forms a trigger, snap through action of the V-Beams actuating the grasper tip. The snap provides an impulse that actuates the gripper and holds its position until reset. The grasper adapts to the tissue geometry, due to its compliant nature, reducing risk of bruising. Our primary design workflow included analyzing the forces transferred due to the V-Beam arrangement, establish its design parameters and come up with the model of the handheld trigger mechanism.

4.1 Introduction to Compliant and Bistable Mechanisms

Compliant mechanisms are flexible mechanical structures that achieve motion through their elastic deformation, and transform the force and displacement at least partly through the deformation of their structural components, enabling morphing actions. The trigger through its compliance, allows the deformation of V-Beams leading to grasping action. Arches are some of the most commonly used structural elements, and when subjected to transverse load, they undergo a non-linear snap through buckling resulting in two force free equilibria. These are known as curved beam bistable mechanisms [2] [3]. Its force displacement characteristics are shown in Figure 1.

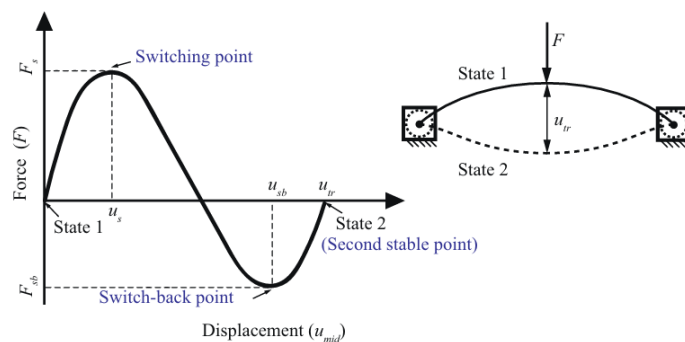


Figure 1: Typical force-displacement characteristics of a bistable arch [2]

State-1 is the un-stressed force free equilibrium state and State-2 is the stressed force free equilibrium state. The zero force state between states 1 and 2 is an unstable equilibrium state at half the travel displacement of the arch. Switching force F_s , is the minimum force required to switch from State 1 to 2, and F_{sb} is the switch back force, the minimum force required to go back to State 2 from 1. The travel u_{tr} is the distance the midpoint of the arch moves between its two stable states. We examined the bistable nature of the V-Beams in our mechanism in a narrow range which is sufficient for the grasper,

and therefore our analysis was focused on the switching forces and the travel only, thus allowing us to perform semi-analytic methods to find these critical points.

4.2 V-Beam Mechanism and its Design Process

The V-Beam is a very common compliant bistable structure, with compliant beams and a shuttle, which is regarded as a slider with only one degree of freedom. When a force greater than its switching force is applied the state transition of the V-Beam will occur as shown in Figure 2, where stable state-1 is the solid line and stable state-2 is the dashed line. Of the methods available to analyze the performance

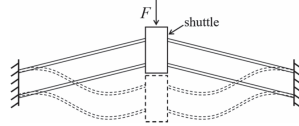


Figure 2: Structure of V-beam mechanism [4]

and parameters of a bistable mechanism, the most relevant methods are the Pseudo-Rigid Body Model (PRBM) and non-linear FEA. Although it has a low accuracy, the PRBM is more suitable for a low-demand analysis and the initial design of the compliance mechanism, and was easy to follow through due to its intuitiveness. Further non-linear FEA was done using ABAQUS (in section 4.3) for verifying the PRBM's predictions, and measure its closeness to the required forces and travel. A direct inference wasn't made from non-linear FEA alone, since these models usually mispredict a V-Beam buckling mode unless a virtual linear perturbation or a constraint force is initially exerted on the shuttle, and the ABAQUS analysis was done for verifying the mechanism's working principles.

4.3 Force and Travel Analysis

An iterative analysis was carried out using the Pseudo-Rigid Body Model (PRBM-TEBCM) equations for a V-Beam, and the predicted travel was subsequently validated through ABAQUS buckling simulations using the derived parameters.

For the ABAQUS analysis, we tested on V-Beams whose width 'w' is negligible compared to its length 'l'. Our initial width was 2.25 mm and length being 135 mm. The thickness of the beams were 7.5 mm, with an inclination 27.5° . The analysis began with a pre-buckling linear perturbation step where reference loads or boundary conditions that represent the actual service conditions of the V-beam are applied (axial compression, lateral load, or combined loading). 5 eigenvalues corresponding to its eigenmodes were extracted for the initial mesh in the Static, Riks analysis, with the first mode chosen as our desired model. The Riks method in ABAQUS is a path-following algorithm designed to capture load-displacement response through points of instability. Riks algorithm increments an arc-length parameter that controls both load and displacement simultaneously. This allows the solver to pass through limit points (where slope of load-displacement curve is zero) and continue into the post-buckling regime. Properties corresponding to PLA were assigned to the material, along with self interaction properties for contact load modelling. A pressure load of $1 \frac{N}{mm^2}$ was applied to the shuttle and the eigenvalues corresponding to the modes were used for applying forces in the Riks Analysis. The analysis was then done for our final parametric model from the TECBM equations (Trial 2) to prove its robust design, with single and parallelly arranged V-Beams. The details for both our trials have been summarized in Table 1 and Table 2.

The first trial with Riks Analysis showed that, while bistability was achieved, travel was excessive and beam dimensions were impractical for a compact medical trigger device. But it highlighted the potential benefit of using multiple V-Beams, in helping reduce beam length at the same time achieve required travel

Table 1: V-beam buckling and travel results (Trial 1). First row corresponds to single V-Beam and second row corresponds to 3 parallel V-Beams. The travel corresponds to λ_1 / eigenmode 1.

Length (mm)	Thickness (mm)	Width (mm)	Angle (deg)	λ_1	Travel (mm)
135	2.25	7.5	27.5	6.684e-2	9.298
135	2.25	7.5	27.5	0.19742	27.125

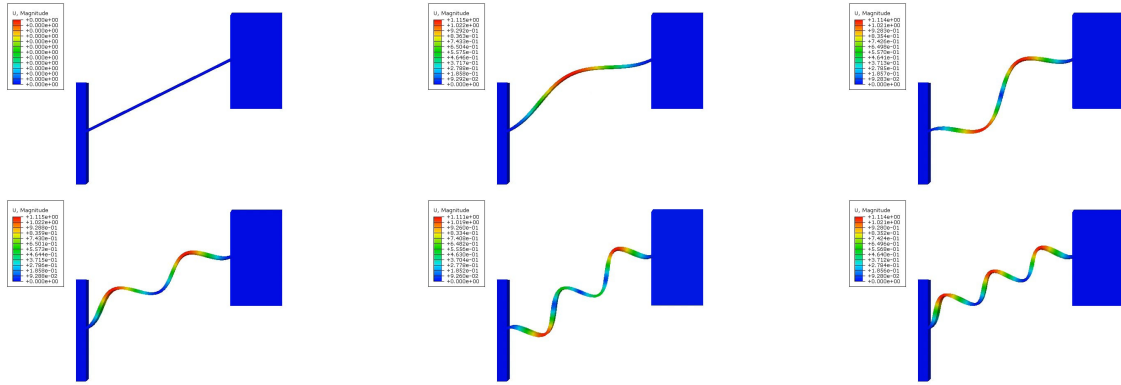


Figure 3: Pre-buckling analysis for eigenmodes (linear perturbations, Trial 1).

and force output. The effective stiffness and snap-through force can be tailored by adjusting the number of beams, their thickness, and spacing. This makes it possible to design a trigger with predictable and safe actuation force. Nevertheless, this initial analysis was valuable because it highlighted a clear design direction: rather than relying on one long V-beam, a configuration using multiple smaller V-beams in parallel or series could achieve the required travel within acceptable size constraints.

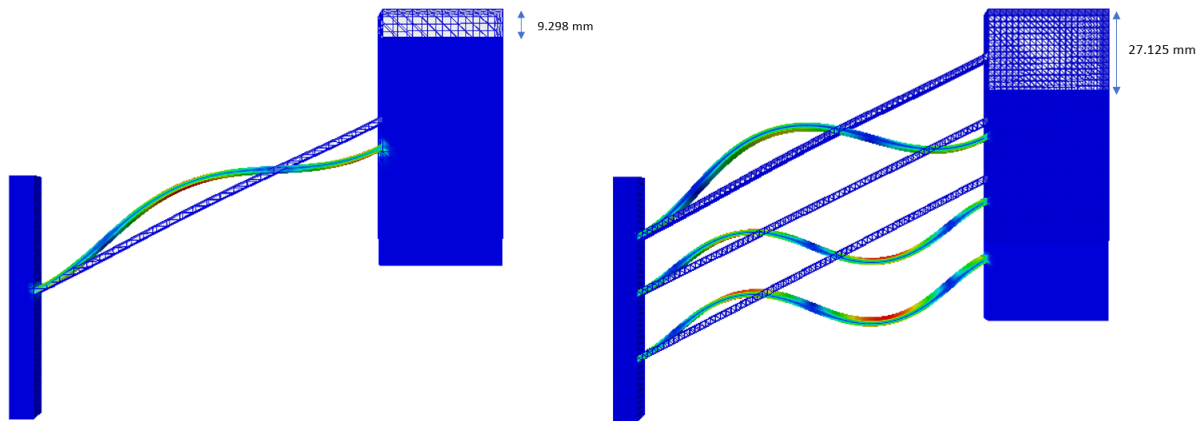


Figure 4: Riks analysis for shuttle travel (Trial 1). The results showed that although the beam successfully demonstrated snap-through behavior, the geometry selected for this trial produced beams that were excessively long. Such dimensions would not be practical for integration into a medical device, since they resulted in disproportionately large travel outputs that cannot be accommodated within a compact trigger mechanism.

In order to estimate the geometry of the laparoscopic gripper as a whole, it is necessary to determine the size and number of beams required for the specific value of force required. Comparing with other laparoscopic grippers in the market, we decided on a nominal value of 30-35N as a starting point for our

prototype [5]. This force would correspond to the end force that the compliant gripper would apply on the tissue. For the preliminary analysis, we consider that the gripper mechanism has a transmission ratio of 1, i.e., it transmits exactly the force it receives.

For the purpose of simplifying calculations, the beams were approximated as bistable single V-beams so that their length, thickness, width, and tilt angle may be approximated for the required force, using the two elements beam constraint model (TEBCM) [4].

The following inequality from Liu, M. et al.[4] is used to determine the set of equations required to obtain the values of the necessary geometric parameters:

$$D_1 = -5.365t^2 - 1.163y_o^2 - 3.257x_o \geq 0 \quad (1)$$

Substituting the values of normalized thickness t , x_o , and y_o , we obtain

$$-4.652 \frac{\cos^2 \theta}{L^2} \Delta_y + 6.514 \frac{\sin \theta}{L} \Delta_y - 21.46 \frac{T^2}{L^2} \geq 0 \quad (2)$$

If the required Δ_y satisfies the inequality, then the following equations are used to calculate f_o and p_o :

$$\begin{aligned} f_o &= f_2 = f_1 = -4.8618y_0 \\ p_o &= p_2 = p_1 = -9.8837 \end{aligned}$$

Else, the following set of equations are used:

$$\begin{aligned} k_1 &= \frac{4t^2}{675} + \frac{y_0^2}{42000} \\ k_2 &= \frac{16t^2}{45} + \frac{17y_0^2}{8100} - \frac{8x_0}{225} \\ k_3 &= \frac{16t^2}{3} - \frac{96y_0^2}{175} + \frac{32x_0}{15} \\ k_4 &= -\frac{48y_0^2}{5} - 32x_0 \\ c_1 &= -k_2^2 + 3k_1k_3 \\ c_2 &= \left[-2k_2^3 + 9k_1k_2k_3 - 27k_1^2k_4 + \sqrt{4c_1^3 + (-2k_2^3 + 9k_1k_2k_3 - 27k_1^2k_4)^2} \right]^{1/3} \\ p_o &= p_1 = -\frac{k_2}{3k_1} + \frac{\sqrt{2}c_1}{3k_1c_2} + \frac{c_2}{3\sqrt{2}k_1} \\ f_o &= \frac{y_0}{2} \left[\left(12 + 6p_1 + \frac{1}{700}p_1^2 \right) \left(\frac{-6 - 10p_1 + \frac{1}{1400}p_1^2}{(4 + \frac{2}{15}p_1 - \frac{11}{6300}p_1^2)} \right)^2 \right] \end{aligned}$$

Once p_o and f_o are obtained, the force required to initiate the buckling mode, F , is calculated using the below equation.

$$F = - \left(\frac{4EI f_o}{L^2} \cos \theta + \frac{4EI p_o}{L^2} \sin \theta \right) \quad (3)$$

E and I are the Young's modulus of the material, PLA, and moment of inertia of the beam cross section respectively. The force obtained corresponds to a double V-beam. In our application, we are using a

single V-beam. Hence, the actual value of force is approximated to be half the value obtained from Equation 3. The geometric parameters obtained from the calculations are summarized in Table 2.

Table 2: Results obtained from calculations using V-beam approximation.

Length (mm)	Thickness (mm)	Width (mm)	Tilt Angle (deg)	Travel (mm)	Force (N)
40	1.2	5	7	5	3.505

With each beam providing about 3.5N force, we settled on using 12 beams for a total force of about 42N, which is slightly higher than our initial target, but still well within the range of conventional laparoscopic grippers.

The following figures demonstrate the results of FEM analysis on a single V-beam with the same geometric parameters as obtained using the TEBCM method. The analysis was carried out to verify the buckling motion of the shuttle.

Table 3: V-beam buckling and travel results (Trial 2). First row corresponds to single V-Beam and second row corresponds to 3 parallel V-Beams. The travel corresponds to λ_1 / eigenmode 1.

Length (mm)	Thickness (mm)	Width (mm)	Angle (deg)	λ_1	Travel (mm)
40	1.2	5	7	0.19587	6.72075
40	1.2	5	7	0.76156	11.4203

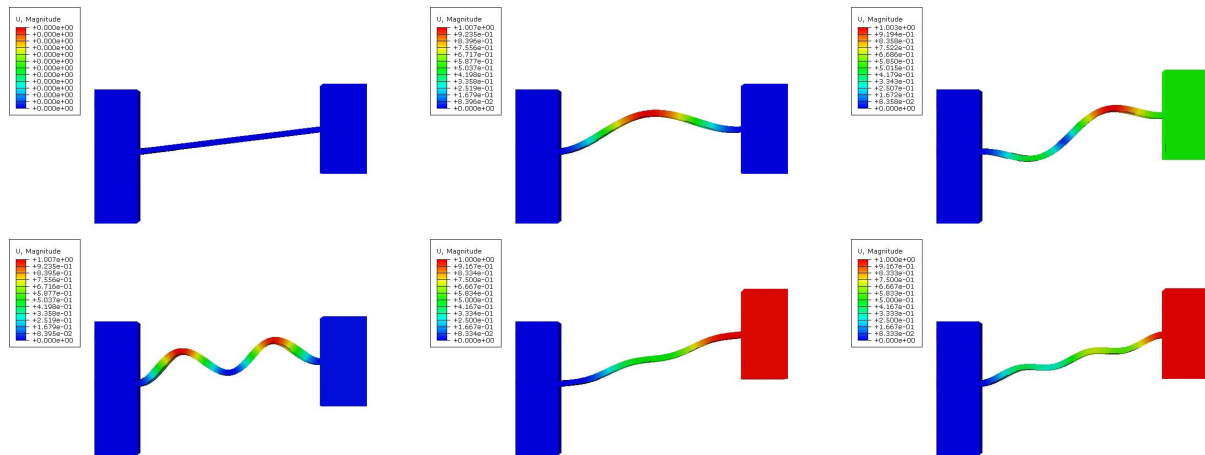


Figure 5: Pre-buckling analysis for eigenmodes (linear perturbations, Trial 2).

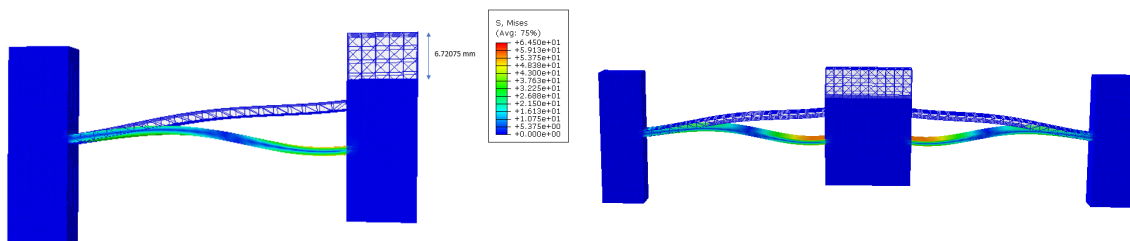


Figure 6: Riks analysis for Trial 2 with parameters from PRBM formulation. Left image shows a symmetric half of the V-Beam and right image shows the complete assembly.

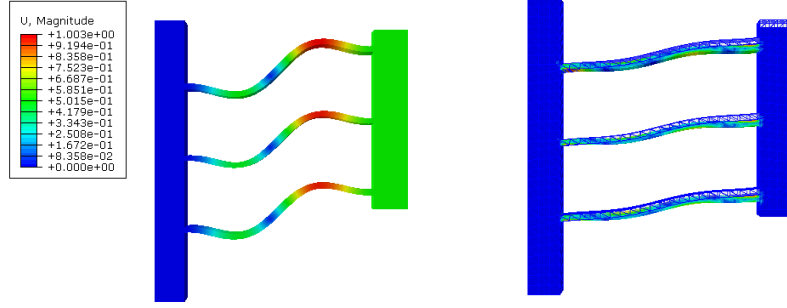


Figure 7: Analysis with Parallel V-Beam arrangement using the derived parameters for Trial 2. The spacing between each V-Beam is 8mm, and the travel and switching force increased due to increase in number of beams.

4.4 Working Principle of Trigger Induced Bistable Mechanism

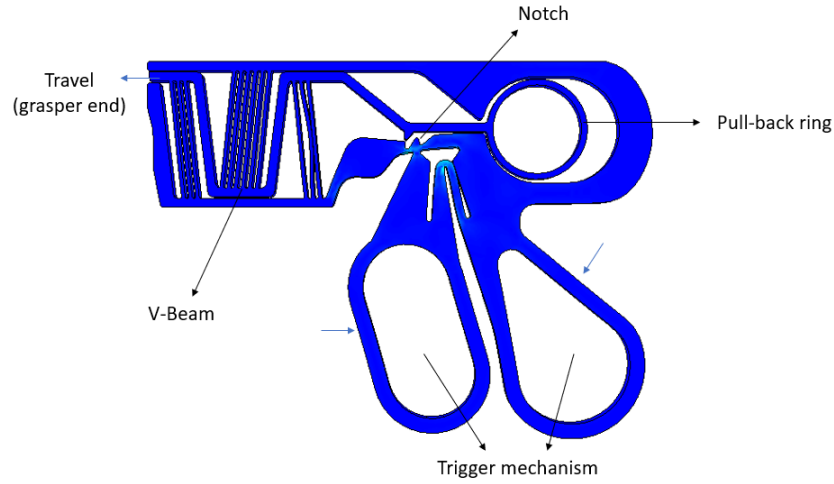


Figure 8: 2D view of the complete Trigger Mechanism in the final unstressed state.

The conceived trigger mechanism, in conjunction with an added compliant gripper mechanism, replaces the conventional laparoscopic gripper by performing all its required functions using compliant mechanisms instead of moving parts. The circular push-back ring is pushed backwards to move the beams into their second state (buckled stressed state), from where it is prevented from moving by a small notch. Once the trigger is actuated, the notch moves downward and the shuttle forward, as the beams return to their unstressed initial state. The forward motion of the shuttle will actuate the compliant gripper, closing it on the required tissue at the time of application. The following ABAQUS analysis shows the bistability of the V-Beams as the push back maneuver happens, as the grasper relaxes its actuation, as the shuttle travels back.

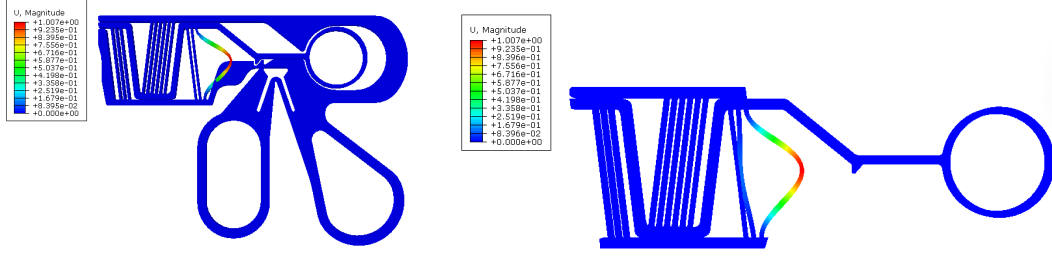


Figure 9: 1st eigenmodes for the full mechanism (left) and the V-Beam arrangement in the grasper (right).

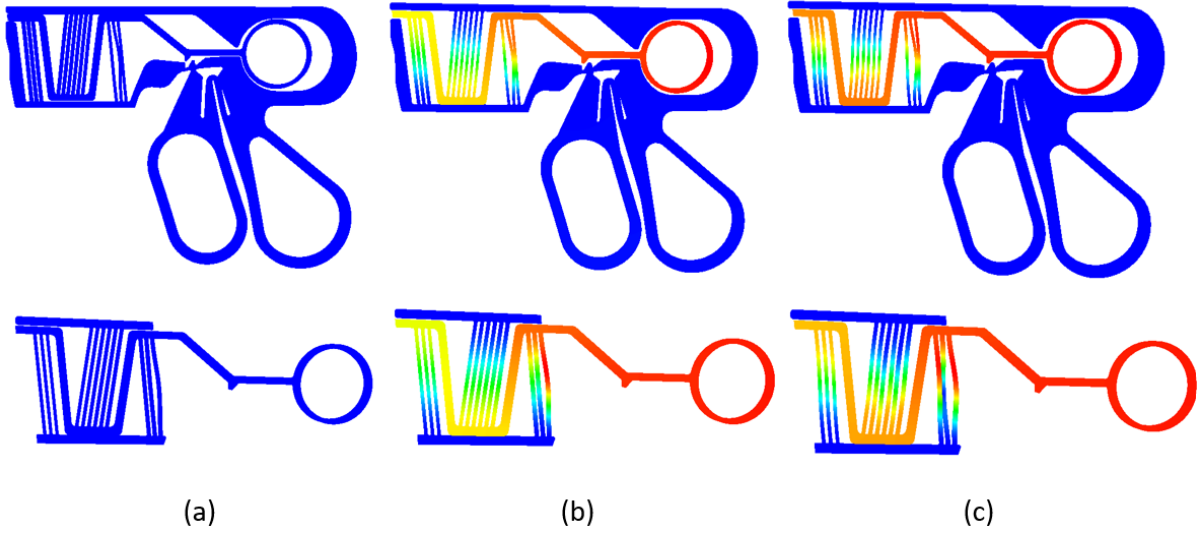


Figure 10: Pull back maneuver of the bistable, trigger induced compliant mechanism, showing the beams snapping to their first eigenmodes progressively from (a) to (c). Top row is the full mechanism, and the bottom row shows only the V-Beam and the pull-back ring.

4.5 Compliant Grasper (Ongoing work)

Once the trigger mechanism is designed, the focus shifts to the compliant grasper, which will serve as the end-effector in this application. While the specific geometry and configuration of the grasper are yet to be finalized, several key design considerations have become evident over the course of the trigger design.

Firstly, the grasper must close when the shuttle moves forward. This introduces an inverse mechanism compared to a conventional grasper which is expected to close when its actuator is retracted. Second, the motion of the grasper must be smooth and controlled. Given the sensitivity of this application, sudden motion and abrupt forces may cause catastrophic results. Hence, the force exerted by the trigger must be transmitted smoothly and gradually by the compliant grasper mechanism.

The force transmission ratio between the grasper and the trigger mechanism must also be optimized. This is important to ensure that the end effector does not exert a force more than the approximated permissible value of 40N on the tissue during the operation. The design of a suitable compliant grasper mechanism, based on these constraints and its full prototyping, will constitute the next phase of the laparoscopic gripper development.

5 Inventiveness, Novelty and Non-Obviousness

The bistable compliant laparoscopic grasper is fundamentally different from other conventional laparoscopic graspers which use combinations of mechanical linkages to achieve the desired motion. The compliant grasper eliminates the complexity involved in the design, manufacture, and assembly of multiple parts, and provides a lightweight alternative to conventional graspers. While design improvement in the field has predominantly focused on achieving better output ratio, this design changes the fundamental working principle of the grasper itself.

6 Comparative Analysis

Table 4: Comparison between Conventional Rigid-Link Grasper and Proposed Compliant V-Beam Bistable Grasper

Parameter	Conventional Model	Proposed Compliant Model
Efficiency	Moderate to high, suffers losses due to moving parts	Comparable to conventional model, but less losses.
Cost	High (due to precision linkages, joints and assembly)	Low (monolithic compliant structure, only 2 parts)
Weight	Comparable (200-350 g)	Comparable
Complexity	High (requires precision machining, multiple moving parts)	Low (uses FDM techniques)
Safety	Risk of tissue damage due to poor force feedback	Better, improved force feedback

7 Impact of the Design

The primary advantage of the novel compliant laparoscopic gripper mechanism is its ability to provide reliable haptic feedback. By ensuring that the compliant grasper end-effector remains closed when the push-back ring is in its neutral state, the mechanism enables the user to intuitively perceive whether or not the tissue is being grasped.

Another significant advantage of the compliant mechanism is its cost-effectiveness. The elimination of multiple discrete components reduces not only material costs but also the expenses associated with assembly and the stringent tolerances typically required for interlinked parts. Furthermore, since the grasper is designed as a planar structure, it can be easily manufactured using additive manufacturing techniques such as 3D printing, or even through extrusion processes for alternative materials.

8 Estimated Pricing

Since the prototype will be fabricated using PLA through 3D printing, the cost of production is primarily determined by the weight of the mechanism. Based on estimates in Autodesk Fusion, the trigger mechanism alone weighs approximately 40 grams, while the compliant grasper is expected to weigh about 20 grams, resulting in a total prototype weight of around 60 grams. Using fused deposition modeling (FDM) with a standard 0.2 mm nozzle, online services quote a fabrication cost of approximately Rs. 700. Alternatively, in-house manufacturing could reduce the cost to as little as Rs. 50, covering only the material cost, with electricity costs being additional.

For a fully functional laparoscopic gripper, however, the mechanism would require sterile coverings to ensure suitability for practical applications. This would introduce additional manufacturing and assembly costs, which will be evaluated during the subsequent stages of gripper development.

9 Possible Risks and its Handling

The most critical risk associated with the design is the potential failure of the part. The compliant mechanism relies on several slender beams, which are inherently more susceptible to fracture. Consequently, the mechanism must be designed with an adequate factor of safety to ensure structural integrity.

A related concern is the reliability of finite element method (FEM) simulations performed in ABAQUS. Given the inherent approximations within FEM, combined with the sensitive and precise nature of this application, even minor discrepancies in the numerical results can translate into significant variations in practice. This limitation is further compounded by the specific requirements of buckling analysis in ABAQUS, where an initial perturbation load must be applied to introduce asymmetry into the structure. Such conditions inevitably influence the outcome of the analysis. Therefore, FEM results should be interpreted only as indicative, while the results obtained from prototype testing must be regarded as definitive.

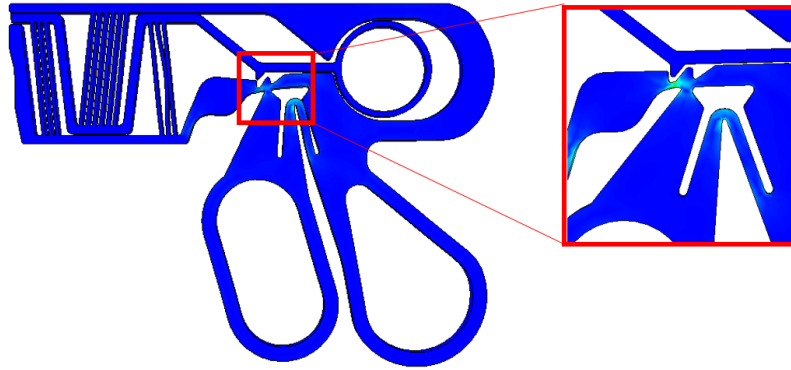


Figure 11: Stress concentrations in thin regions in the mechanism. The picture is taken from a failed ABAQUS analysis. The support structures around the notch are too compliant, leading to failure during operation. This requires a redesign of these areas.

10 Acknowledgments

We would like to acknowledge the Compliant Mechanisms and Robotics Group at Brigham Young University and the Nano Access Lab at TU Eindhoven, whose Maker resources helped us immensely in the initial design phase. We would also like to thank the CAE Lab at Indian Institute of Technology, Hyderabad, for providing access to Solidworks and ABAQUS for the modeling and analysis of the mechanism.

References

- [1] Dobbelsteen, J., Lee, R., van Noorden, M. & Dankelman, J. Indirect measurement of pinch and pull forces at the shaft of laparoscopic graspers. *Medical biological engineering computing* **50**, 215–21 (2012).
- [2] Palathingal, S. & Ananthasuresh, G. Design of bistable arches by determining critical points in the force-displacement characteristic. *Mechanism and Machine Theory* **117**, Pages 175–188 (2017).
- [3] Qiu, J., Lang, J. & Slocum, A. A curved-beam bistable mechanism. *Journal of Microelectromechanical Systems* **13**, 137–146 (2004).
- [4] Liu, M. *et al.* A bistable criterion for the v-beam mechanism. In *2021 IEEE 16th International Conference on Nano/Micro Engineered and Molecular Systems (NEMS)*, 1106–1109 (2021).
- [5] Olig, E. M., Wilson, S. & Reddy, M. Output force and ratio of laparoscopic graspers: an evaluation of operating room ergonomics. *American Journal of Obstetrics and Gynecology* **229**, 307.e1–307.e9 (2023). URL <https://www.sciencedirect.com/science/article/pii/S0002937823003149>.